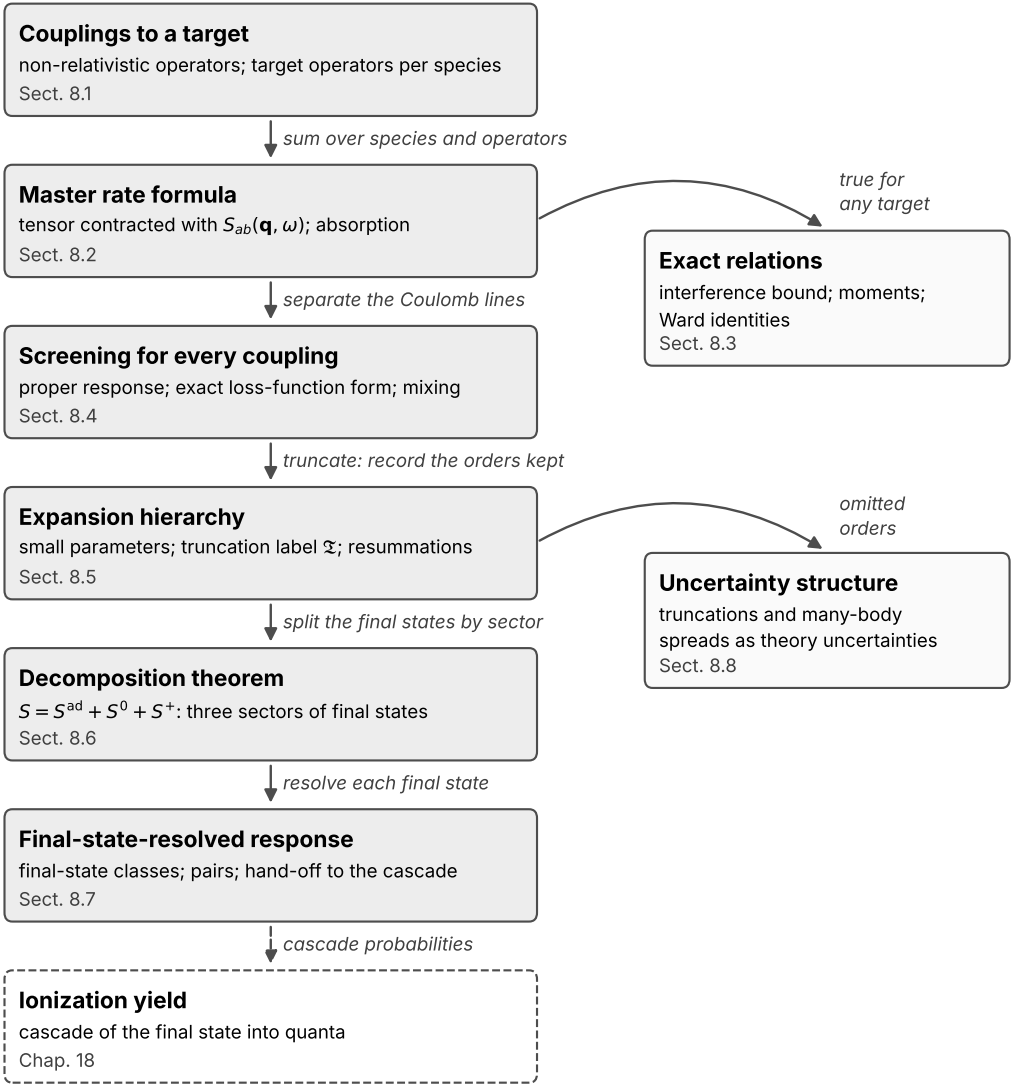
Couplings to a target
non-relativistic operators; target operators per species
Sect. 8.1
sum over species and operators
Master rate formula
tensor contracted with $S_{ab}(\mathbf{q}, \omega)$; absorption
Sect. 8.2
true for any target
Exact relations
interference bound; moments; Ward identities
Sect. 8.3
separate the Coulomb lines
Screening for every coupling
proper response; exact loss-function form; mixing
Sect. 8.4
truncate: record the orders kept
Expansion hierarchy
small parameters; truncation label $\mathfrak{T}$; resummations
Sect. 8.5
omitted orders
Uncertainty structure
truncations and many-body spreads as theory uncertainties
Sect. 8.8
split the final states by sector
Decomposition theorem
$S = S^{\mathrm{ad}} + S^{0} + S^{+}$: three sectors of final states
Sect. 8.6
resolve each final state
Final-state-resolved response
final-state classes; pairs; hand-off to the cascade
Sect. 8.7
cascade probabilities
Ionization yield
cascade of the final state into quanta
Chap. 18